

Question 2

	C_1	C_2	C_3
Red	4	2	2
Blue	3	5	2
Purple	3	2	4

Assign each cluster to its majority class:

$$C_1 \rightarrow \text{Red}, \quad C_2 \rightarrow \text{Blue}, \quad C_3 \rightarrow \text{Purple}.$$

Red class:

$$P_R = \frac{4}{10}, \quad R_R = \frac{4}{8},$$
$$F_R = \frac{2P_R R_R}{P_R + R_R} = \frac{2 \cdot \frac{4}{10} \cdot \frac{4}{8}}{\frac{4}{10} + \frac{4}{8}} = \frac{4}{9}.$$

Blue class:

$$P_B = \frac{5}{9}, \quad R_B = \frac{5}{10},$$
$$F_B = \frac{2P_B R_B}{P_B + R_B} = \frac{10}{19}.$$

Purple class:

$$P_P = \frac{4}{8}, \quad R_P = \frac{4}{9},$$
$$F_P = \frac{2P_P R_P}{P_P + R_P} = \frac{8}{17}.$$

Macro-averaged Precision:

$$P_{\text{macro}} = \frac{1}{3} \left(\frac{4}{10} + \frac{5}{9} + \frac{4}{8} \right) = \frac{131}{270} \approx 0.485.$$

Macro-averaged Recall:

$$R_{\text{macro}} = \frac{1}{3} \left(\frac{4}{8} + \frac{5}{10} + \frac{4}{9} \right) = \frac{25}{54} \approx 0.481.$$

Macro-averaged F-score:

$$F_{\text{macro}} = \frac{1}{3} \left(\frac{4}{9} + \frac{10}{19} + \frac{8}{17} \right) = \frac{4190}{8721} \approx 0.480.$$

Question 3

We use the B-CUBED definitions:

$$P_{B^3} = \frac{1}{N} \sum_i \frac{|\text{cluster}(i) \cap \text{class}(i)|}{|\text{cluster}(i)|},$$

$$R_{B^3} = \frac{1}{N} \sum_i \frac{|\text{cluster}(i) \cap \text{class}(i)|}{|\text{class}(i)|}.$$

From Figure 1, the contingency table is:

	C_1	C_2	C_3	Total
Red	4	2	2	8
Blue	3	5	2	10
Purple	3	2	4	9
Total	10	9	8	27

B-CUBED Precision

$$P_{B^3} = \frac{1}{27} \left(\frac{4^2 + 3^2 + 3^2}{10} + \frac{2^2 + 5^2 + 2^2}{9} + \frac{2^2 + 2^2 + 4^2}{8} \right) \quad (1)$$

$$= \frac{1}{27} \left(\frac{34}{10} + \frac{33}{9} + \frac{24}{8} \right) \quad (2)$$

$$= \frac{1}{27} (3.4 + 3.6667 + 3) \quad (3)$$

$$= \frac{10.0667}{27} \approx 0.373. \quad (4)$$

B-CUBED Recall

$$R_{B^3} = \frac{1}{27} \left(\frac{4^2 + 2^2 + 2^2}{8} + \frac{3^2 + 5^2 + 2^2}{10} + \frac{3^2 + 2^2 + 4^2}{9} \right) \quad (5)$$

$$= \frac{1}{27} \left(\frac{24}{8} + \frac{38}{10} + \frac{29}{9} \right) \quad (6)$$

$$= \frac{1}{27} (3 + 3.8 + 3.2222) \quad (7)$$

$$= \frac{10.0222}{27} \approx 0.371. \quad (8)$$

B-CUBED F-score

$$F_{B^3} = \frac{2P_{B^3}R_{B^3}}{P_{B^3} + R_{B^3}} \quad (9)$$

$$\approx 0.372. \quad (10)$$

$P_{B^3} \approx 0.373, R_{B^3} \approx 0.371, F_{B^3} \approx 0.372$